Ciprian Adrian **Corneanu**

Scientist - Machine Learning and Computer Vision

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Professional Summary: *Seasoned Computer Vision researcher and developer with over a decade of expertise in advancing state-of-the-art AI technologies. Specialized in Deep Learning architectures, with particular focus on Human Analytics, Generative AI and Vision-Language integration. Demonstrated research excellence through multiple publications in premier venues including CVPR, ECCV, and IEEE TPAMI. Recognition for scientific impact includes CVPR 2020 Best Paper Award nomination. Proven track record in bridging theoretical innovations with practical applications.*

Experience

Senior Applied Scientist
Seattle, USA

Amazon — Ads, advertising technology at scale. *Mar. 2025 - Now*

- Architecting agentic frameworks that orchestrate generative models for end-to-end video ad creation through natural language.

Applied Scientist
Seattle, USA

Amazon — Stores, North American e-commerce. *Feb. 2022 - Feb. 2025*

- Developed automated workflows for curating and maintaining ultra-large-scale multimodal datasets(+1B).
- AI model training on massive GPU clusters.

Applied Scientist
Munich, Germany

Tawny - Start-up focused on Human Analytics. *Feb. 2020 - Jan. 2022*

- Spearheaded R&D initiatives in Human Analytics, driving innovations in emotion recognition and behavioral understanding.
- Architected and supervised the development of a comprehensive facial expression dataset.
- Successfully delivered production-ready facial analysis pipeline.

Development Engineer
Freiburg/Villingen, Germany

C.R.S. iiMotion. Custom designs for video processing and electronics. *Oct. 2011 - Sep. 2014*

- Developed and optimized real-time optical flow algorithms for plasma TV motion enhancement. Successfully implemented on FPGA hardware and deployed across LG Electronics' consumer TV product line.
- Contributed to comprehensive optical measurement solution for medical endoscopes, delivering end-to-end C# software suite for Schoelly, enabling precise quality control in medical device manufacturing.

Research Experience

Research Scholar
Barcelona, Spain

University of Barcelona, Computer Vision Center *Feb. 2015 - Dec. 2019*

- Improving automatic facial expression analysis using deep artificial neural networks.

Visiting Scholar
Columbus, USA

Ohio State University *Sep. 2018 - Mar. 2019*

- Using topological descriptors for analyzing deep artificial neural networks.
- Advisor: Aleix Martinez

Intern
Villingen, Germany

CRS iiMotion *Feb. 2011 - Sep. 2011*

- Implementing optical flow algorithms for frame rate conversion on plasma TVs.

Education

Ph. D. in Computer Science
Barcelona, Spain

University of Barcelona *2015 - 2019*

- Thesis: *Improving Performance and Interpretability in Recognizing Facial Action Units with Deep Neural Networks*
- Advisor: Sergio Escalera

M.Sc. in Computer Vision

Barcelona, Spain

AUTONOMOUS UNIVERSITY OF BARCELONA

2014 - 2015

- Thesis: *Facial Expression Analysis of Neurologically Impaired Children*
- Advisor: Sergio Escalera

M.Sc. in Telecommunications

Paris, France

TÉLÉCOM SUDPARIS

2009 - 2011

- Thesis: *Motion Estimation. Dealing with Covered and Uncovered Regions.*
- Advisor: Nicolas Rougon, External Advisor: Carlos Correa

B.S. in Electronics, Telecommunications and Information Technology

Bucharest, Romania

TECHNICAL UNIVERSITY OF BUCHAREST

2006 - 2009

Publications

CONFERENCES

- *Be More Specific: Evaluating Object-centric Realism in Synthetic Images. Anqi Liang, Ciprian Corneanu, Qianli Feng, Giorgio Giannone, Aleix Martinez, Proceedings of the IEEE Conference on Computer Vision (CVPR-under rev.), 2025*
- *Structured human assessment of text-to-image generative models. Ciprian Corneanu, Qianli Feng and Aleix Martinez. Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025*
- *LatentPaint: Image Inpainting in Latent Space with Diffusion Models. Ciprian Corneanu, Raghudeep Gadde, Aleix M. Martinez; Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024*
- *Computing the Testing Error without a Testing Set. Corneanu, Ciprian and Madadi, Meysam and Escalera, Sergio and Martinez, Aleix, Proceedings of the IEEE Conference on Computer Vision (CVPR), 2020*
- *What Does It Mean to Learn in Deep Networks? And, How Does One Detect Adversarial Attacks?. Ciprian Corneanu, Meysam Madadi, Sergio Escalera, Aleix Martinez. Proceedings of the IEEE Conference on Computer Vision (CVPR), 2019*
- *Deep structure inference network for facial action unit recognition. Ciprian Corneanu, Meysam Madadi, Sergio Escalera. Proceedings of the European Conference on Computer Vision (ECCV), 2018*
- *Continous Supervised Descent Method for Facial Landmark Localisation. Marc Oliu Simon, Ciprian Corneanu, Laszlo Jeni, Jeffrey Cohn, Takeo Kanade. Asian Conference on Computer Vision (ACCV), 2016*

JOURNALS

- *Rubén Ballester, Xavier Arnal Clemente, Carles Casacuberta, Meysam Madadi, Ciprian A. Corneanu, Sergio Escalera, Predicting the generalization gap in neural networks using topological data analysis, Neurocomputing, Volume 596, 2024*
- *Automatic Recognition of Deceptive Facial Expressions of Emotion. Ikechukwu Ofodile, Kaustubh Kulkarni, Ciprian Adrian Corneanu, Sergio Escalera, Xavier Baro, Sylwia Hyniewska, Juri Allik, Gholamreza Anbarjafari. Transactios in Affective Computing (TAC), 2017*
- *Corneanu, C. A., Oliu, M., Cohn, J. F., & Escalera, S. . Survey on RGB, 3D, Thermal, and Multimodal Approaches for Facial Expression Recognition: History, Trends, and Affect-related Applications. Transactions in Pattern Analysis and Machine Intelligence (TPAMI), 2016*
- *Marc Oliu Simon, Ciprian Corneanu, Kamal Nasrollahi, Sergio Escalera Guerrero, Olegs Nikisins, Yunlian Sun, Haiqing Li, Zhenan Sun, Thomas B Moeslund, Modris Greitans. Improved RGB-DT based face recognition. Iet Biometrics, 2016*

See ***Google Scholar*** *for a complete list.*

Honours & Awards

- The paper *Continous Supervised Descent Method for Facial Landmark Localisation. Marc Oliu Simon, Ciprian Corneanu, Laszlo Jeni, Jeffrey Cohn, Takeo Kanade. Asian Conference on Computer Vision, 2016* got accepted for an oral presentation. The acceptance rate for oral presentations at the Asian Conference of Computer Vision for the year 2014, was 3.9%.
- The paper *Computing the Testing Error without a Testing Set. Ciprian Corneanu, and Meysam Madadi, Sergio Escalera and Aleix Martinez (CVPR, 2020), was Best Paper Award Nominee. CVPR is the most important Computer Vision conference of the year and the 2020 edition got a record number of submissions (>6000). Only 20 papers were nominated for Best Paper.*

Organization

ChaLearn Looking at People 2016 ECCV Workshop and Challenge

2016

Amsterdam, The Netherlands

ChaLearn Looking at People 2016 CVPR Workshop and Challenge

2016

Las Vegas, USA

Invited Talks

The Topology of Learning *2020*
At Technical University of Munich (group of Laura Leal-Taixé), University of Toronto (group of Sanja Fidler), Carnegie Mellon University (group of Fernando de la Torre)

Prediction Testing Error without a Testing Set *2020*
At Computer Vision Center, Barcelona

What does it mean to learn in Deep Neural Networks *2019*
At BCN-AI, Barcelona

Introduction to Affective Computing and Facial Expression Analysis *2017-2018*
At Data Science MSc Program, University of Barcelona

Undergraduate Supervision

Increasing Robustness of Facial Expression Recognition against Speech *2021*
Degree: M. Sc. Student: Marcel Emre Baur, Ludvig-Maximilians University, Munich, Germany.

An Analysis of Metric and Topological Estimators of Generalization in Deep Learning *2021*
Degree: B.S. Student: Xavi Arnal, University of Barcelona, Barcelona, Spain.

Topological Descriptors in Deep Learning and their Connection with Model Generalization *2021*
Degree: B.S. Student: Ruben Ballester Bautista, University of Barcelona, Barcelona, Spain.

Facial Analysis in the Context of Human-Machine Interactions. *2015*
Student: Aitor Moreso, University of Barcelona, Barcelona, Spain.